

Assignment - 7  
MDP Formulation

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10) Calculate Undiscounted & Discounted Returns.

Given;

$$R_0 = 2 ; R_1 = 3, R_2 = 4$$

and

$$\gamma = 0.9$$

\* Undiscounted Return

$$G = R_0 + R_1 + R_2$$

$$= 2 + 3 + 4 \Rightarrow 9$$

\* Discounted Return

$$G = R_0 + \gamma R_1 + \gamma^2 R_2$$

$$= 2 + (0.9)(3) + (0.9)^2(4)$$

$$= 2 + 2.7 + 3.24$$

$$\therefore \boxed{G = 7.94}$$

$\therefore$  Undiscounted Return = 9

$\therefore$  Discounted Return = 7.94

209)  $2 \times 3$  Grid World

|   | 0  | 1  | 2             |
|---|----|----|---------------|
| 0 | S0 | S1 | <del>S2</del> |
| 1 | X  | S3 | S4            |

States: S0, S1, S3, S4, G

Actions: Right, Left, Up, Down

Normal Reward = -1

Goal Reward = +10

Hit Wall = -5

Discount factor = 0.9

Suppose  $S_0 \xrightarrow{R} S_1$

then,  $S_1 \xrightarrow{R} G$

$R_0 = -1$ ;  $R_1 = +10$ ,  $\gamma = 0.9$

$$\gamma \quad G = R_0 + \gamma R_1$$

$$h = -1 + (0.9)(10)$$

$$h = -1 + 9$$

$$\therefore \boxed{h = 8} \quad //$$